

standard which sets the parameters and details of how the data segments or packets are routed across different networks. The TCP component of the protocol is required to support the transmission and communication between the different sources. The TCP works as a medium through which different networks can understand each other and communicate.

Another critical piece of this network naming protocol is the device's Media Access Control address or MAC address. In fact, the MAC address is the essential foundation on which IP addresses work over ethernet. Each device on a network that is transmitting and receiving data has a MAC address. The network chip or card used by devices are assigned unique MAC addresses and are the unique identification number of the transmitting device. The IP protocol is layered on top of the ethernet network where MAC address takes predominance. While the IP addresses are assigned by a router or the Internet Service Provider (ISP), the MAC address is hardwired on to the device which makes it a virtual serial number. All the devices that would make up the Internet of Things would require a MAC number which is expressed in the following alpha numeric format - 1a:2b:3c:4d:5e:6f, where each set is a representation of a binary standard. This format uses a 48 bit system which gives us 281, 474, 976, 710, 656 or 2^{48} unique possibilities.

To illustrate this relationship between the protocols and devices let us imagine a work/office computer relationship. The goal is to copy a document from the office PC to the home PC across the network. As the copy is initiated on the office PC, the Transmission Control Protocol or TCP creates a link between the two computers and authenticates the MAC addresses of the devices, and the Internet Protocol defines the routing rules which then connects the different input/output ports. Since TCP has set up the pathway between the two computers it also labels the nature of the data and the size of the file while the IP works to segment the data of the document into packets and labels it with the appropriate address and header data. This ensures that not only does the file reach its source but also the order in which all the packets need to be reorganised to become the required document. The IP does the job of sending across the data and the information in a standardised format while the TCP works to reinterpret it into the original document format which is a single file.

The system of these protocols is essential for devices on a network to communicate with one another. This makes it critical that the TCP/IP information assigned over the MAC address on each device is in the proper configuration. This is done by assigning each device a specific address which